Natural Language Processing

Lecture 2
Core Concepts

Learning

Search

Representations

COURSE: NATURAL LANGUAGE PROCESSING | Dr. Tamal Ghosh

Learning Objectives

01 Explain different learning paradigms in NLP.

02 Understand how machines learn language.

03 Describe search techniques used in NLP.

04 Explain different methods of text representation.

05 Understand the relationship between learning and representation.

06 Prepare for classical NLP models such as Naïve Bayes and Logistic Regression.

From Language to Intelligence

Humans learn via

Reading

Listening

Experience

Machines learn via

Data

Algorithms

Optimization

THE CORE GOAL

Learn a function that maps text to useful outputs.
Examples: Email → Spam | Review → Sentiment | Question → Answer

What Does "Learning" Mean?

Learning is the process of improving performance by analyzing examples.

Instead of writing rules manually, **Machine Learning learns patterns automatically.**

Example Training Data (Email)	Label
"Win ₹10,000 now!"	Spam
"Meeting tomorrow at 10 AM"	Ham
"Congratulations! You've won..."	Spam
"Project report attached for review"	Ham

Types of Learning (Part 1)

SUPERVISED

Training data contains explicit labels for every input.

Spam detection

Sentiment analysis

Document classification

INPUT → LABEL

UNSUPERVISED

No labels available. The machine finds hidden structures.

Topic discovery

Document clustering

Anomaly detection

INPUT → PATTERN

Types of Learning (Part 2)

REINFORCEMENT LEARNING

Learning through a system of **rewards** and **penalties** to achieve an optimal goal.

01

Dialogue Systems

02

Interactive Assistants

03

Robotics & Control

Supervised Learning Workflow

01 Training Dataset → 02 Feature Extraction → 03 ML Algorithm → 04 Model Creation → 05 Prediction & Evaluation

Real-world Example

Movie Review → Feature Extraction → Classifier → Positive/Negative Prediction

Supervised vs Unsupervised

Feature	Supervised Learning	Unsupervised Learning
Data	Uses Labeled Data	No Labels Provided
Goal	Predict Outputs	Find Hidden Patterns
Tasks	Classification, Regression	Clustering, Topic Modeling
Examples	Spam Detection, Sentiment Analysis	Customer Segmentation, Topic Discovery

What is Search in NLP?

> Search means **exploring possible solutions** within a vast space to identify the **best candidate**.

- Machine Translation
- Speech Recognition
- Text Generation
- Spell Checking

GOAL: CHOOSE THE MOST PROBABLE OUTPUT.

Example: Spell Correction

USER INPUT

recieve

receive

recipe

receiver

revive

Search algorithms evaluate all possibilities in the search space and choose the most likely word based on **context** and **probability**.

Search Space Complexity

Input Sentence:

"I love AI"

Possible Translations

Candidate 1: [Sequence A]

Candidate 2: [Sequence B]

Candidate 3: [Sequence C]

... and thousands more

The Challenge

As sentence length increases, the number of possible word combinations grows exponentially.

Search algorithms efficiently identify the BEST CANDIDATE among millions of potential sequences.

Common Search Algorithms

CLASSICAL NLP

Greedy Search

Beam Search

BFS / DFS

Dynamic Programming

A* Search

MODERN LLM DECODING

Greedy Decoding

Beam Search

Top-k Sampling

Top-p (Nucleus) Sampling

Modern models prioritize diversity and quality through probabilistic sampling.

What is Representation?

Computers cannot understand words. They understand only **numbers**.

Raw Text → Numeric Representation → Machine Learning

> *"Good representations improve model accuracy by capturing the underlying semantic structure of language."*

Why Representation Matters

SENTENCE A

"I love NLP."

SENTENCE B

"I enjoy Natural Language Processing."

The Semantic Gap

Humans easily recognize these have similar meanings. However, a computer sees them as entirely different sequences of characters. **Vectors** bridge this gap by mapping meaning to mathematical space.

Types of Text Representation

SYMBOLIC	STATISTICAL	DENSE
Grammar Rules	Bag of Words (BoW)	Word2Vec / GloVe
Dictionaries	TF-IDF	FastText
Parse Trees	N-gram Models	BERT Embeddings
Ontologies		Transformer Vectors

Modern NLP heavily relies on Dense Representations for capturing context.

Example: Bag-of-Words

INPUT SENTENCE

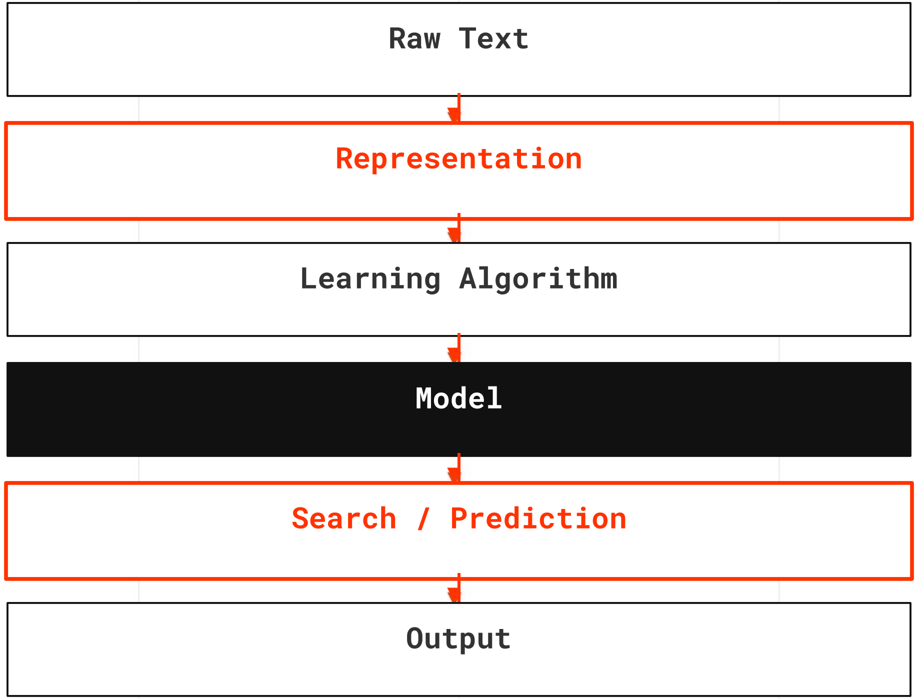
"Cats chase mice."

Cats	Dogs	Chase	Mice	Eat
1	0	1	1	0

[1, 0, 1, 1, 0]

Each position in the vector indicates whether a specific vocabulary word appears in the text.

The NLP Pipeline



Interdependence: Each stage depends on the quality of the previous one. A poor representation will lead to a poor model, regardless of the learning algorithm.

Case Study: Spam Detection

INCOMING EMAIL

"Congratulations! You won ₹5,00,000."

01. Representation

Converts raw text into numeric vectors for processing.

02. Learning

Algorithm identifies patterns associated with "Spam".

03. Search

Selects the most probable label from the output space.

PREDICTION: **SPAM**

Summary and Quiz

LECTURE RECAP

Learning in NLP (Supervised, Unsupervised, RL)

Search strategies and decoding techniques

Text representation and vector space models

The end-to-end NLP pipeline

NEXT LECTURE: BAG-OF-WORDS AND VECTOR SPACE MODELS

IN-CLASS QUIZ

Which learning method requires labeled data?

A. Reinforcement Learning

B. Supervised Learning